

Fraud Transaction Detection using Machine Learning

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Oasis Infobyte – Data Analytics Internship

Level 2 • Task 3

1. Objective

Build a machine learning pipeline to detect fraudulent financial transactions from a highly imbalanced dataset while handling class imbalance using SMOTE. Compare Logistic Regression and Random Forest and evaluate using appropriate classification metrics.

2. Dataset Overview

The dataset contains 10,000 financial transactions with 9,849 legitimate and 151 fraudulent transactions (1.51% fraud rate). Features include transaction amount, hour, merchant category, foreign transaction indicator, location mismatch, device trust score, transaction velocity, and cardholder age.

3. Methodology

- Performed Exploratory Data Analysis to study class imbalance, transaction amounts, and hourly fraud patterns.
- Applied preprocessing using ColumnTransformer with One-Hot Encoding for categorical features and median imputation for numerical features.
- Split the data using stratified sampling.
- Applied SMOTE only to the training data to balance fraud and non-fraud classes.
- Trained Logistic Regression and Random Forest models using an imbalanced-learn pipeline.

4. Model Evaluation

Model	Accuracy	Precision	Recall	F1
Logistic Regression	97%	30%	93%	46%
Random Forest	99%	94%	57%	71%

5. Key Findings

Logistic Regression achieved very high Recall, detecting most fraudulent transactions but producing more false positives. Random Forest achieved much higher Precision and the best F1-score, providing a better balance between fraud detection and false alarms. The ROC-AUC curve indicated excellent discrimination between fraudulent and legitimate transactions.

6. Feature Importance

Feature importance and coefficient analysis showed that variables such as device trust score, foreign transaction status, transaction amount, and transaction velocity were among the most influential predictors. Such insights help financial institutions understand which transaction characteristics contribute most to fraud risk.

7. Why Accuracy is Misleading

Because only 1.51% of transactions are fraudulent, a model predicting every transaction as legitimate would still achieve around 98.5% accuracy. Therefore, Precision, Recall, F1-score, and ROC-AUC provide a more realistic evaluation for fraud detection.

8. Scalability

To support over one million transactions per hour, the solution can be deployed using streaming platforms such as Apache Kafka, distributed processing frameworks like Apache Spark, cloud infrastructure, model APIs, automated monitoring, and periodic retraining.

9. Conclusion

This project demonstrated an end-to-end fraud detection workflow using machine learning. SMOTE effectively addressed class imbalance during training, while the comparison of Logistic Regression and Random Forest highlighted the trade-off between Recall and Precision. Overall, the project strengthened practical skills in preprocessing, handling imbalanced datasets, pipeline creation, model evaluation, and business-oriented interpretation of fraud detection systems.